Modeling Bread Baking with Focus on Overall Deformation and Local Porosity Evolution

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A two dimensional model of bread baking was developed including, for the first time, the dependence of dough viscosity on both temperature and moisture content, the carbon dioxide dissolved from liquid water together with gas generation from yeast at the beginning of baking and the shrinkage due to dough drying. Particular attention was paid to experimental validation of both overall and local variables such as local temperature, overall mass loss, and local moisture content, overall CO_2 released into the oven, and overall deformation and local expansion or shrinkage. Sensitivity studies on generation of carbon dioxide, gravity, and shrinkage are presented to discuss their influences on bread geometry, porosity (reflecting the alveolar structure) and gas pressure. © 2016 American Institute of Chemical Engineers *AIChE J*, 62: 3847–3863, 2016

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Introduction

Modeling of bread baking has improved consistently during recent decades.[1–16] Some models have focused more on the interaction between the oven and the bakery products. Others have proposed refined descriptions of transport phenomena within the product, initially focusing on heat and mass transport, and more recently on their joint contribution to the deformation of gas bubbles.[2,3,6] The scale of experimental verification still remains high, compared to that involved in the set of model equations. This point warrants refinement, since porosity directly affects the sensorial features of the final product (crumb lightness, in-mouth texture[17]) that are important in the purchasing choice and are greatly affected by the processing conditions themselves. For instance, simulations performed by Zhang and Datta[6] showed much lower expansion with higher oven temperature and with microwave processing. In accordance with the aims of this study, the literature analyzed below focuses principally on the development of the alveolar structure in bread dough during baking, although several studies related to bread proving or baking of other cereal products were used to support the analysis because references related to bread baking were too few or nonexistent.

All experimental studies dedicated to bread baking reported a rapid increase in overall volume at the beginning of baking, with maximum inflation in the order of 30%,[18–20] and even higher in specific conditions.[21] Increase in gas pressure is the driving force to explain bread expansion.[2,3,6] Bread dough initially includes unconnected gas bubbles mainly composed of carbon dioxide[22] generated by yeast during proving. The major contribution to overpressure and hence bubble inflation is the desolubilization of carbon dioxide from the liquid phase where it accumulated during proving to the gaseous phase.[23] Production of CO_2 by yeast is also assumed to persist to some extent early in the baking process, before the yeasts are thermally neutralized.[23] To our knowledge, modeling studies have considered dissolved carbon dioxide in the liquid phase only for the proving step.[22,24] Modeling studies for baking have applied a term of CO_2 production directly to the mass balance of the gaseous phase, thus ignoring transport phenomena through the liquid phase.[3,6] Bloksma suggested that water vapor has a moderate role in inflation because water vapor pressure is low between 20 and 60°C. Finally, the increase in temperature in closed bubbles contributes to bubble inflation through the ideal gas law, although to a minor extent[23] ($\sim$10%). After a while, gas bubbles become interconnected, which has the effect of limiting the gas pressure level.[3]

The extent of inflation of gas contained in the bubbles depends on the balance between the pressure forces discussed above and the mechanical resistance of the dough film surrounding the bubbles. In early modeling studies, the mechanical properties of dough were considered constant.[1] More recent studies have used temperature dependence of mechanical properties, with an increase in values within the range of temperature expected for starch gelatinization/protein denaturation.[2,3,6] These were based on previous reports of bread

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Table 1. Table Comparing this Study with Zhang et al. (2005), Lucas et al. (2015), and Nicolas et al. (2014)

Physical parameter	Zhang et al. (2005)	Nicolas et al. (2014)	Lucas et al. (2015)	Present study
Carbon dioxide	• Gas phase • Temperature dependent production	• Gas phase • Temperature dependent production	• Gas phase • Liquid phase desolubilization	• Gas phase • Temperature dependent production • Liquid phase desolubilization
Air	No	No	Gas phase	Gas phase
Mechanical properties	Viscoelastic • Relaxation time = f(T) for $T>T_{gelatinization}$ • Elastic modulus = f(T) ×300 in crust	Viscoelastic • Relaxation time=f(T) ×10 for $T>_{Tgelatinization}$ ×1000 in crust • Elastic modulus=f(T)	Viscous eta = f(T)eta ×1000 for $T>T_{gelatinization}$	Viscous eta = f1(T)*f2(T)*f3(X_w) • f1 influence of T without transformation • f2 influence of transformation stage (gelatinization) • f3 influence of water content
Gravity	Yes	No	No	Yes
Shrinkage due to water loss	No	No	No	Yes
Experimental validation of for local porosity	No	No	Yes	Yes
Pore opening phenomenon	No	Temperature dependent	Temperature dependent	Temperature dependent
Evapo-condensation	Connected pores only	Connected pores only	Connected and unconnected pores	Connected and unconnected pores
Dimension	2D	2D	1D	2D
Oven modeling	No	Yes	No	No

dough viscosity as a function of temperature, although few in number.[23,25,26] However, some important features, such as decreases reported below 60°C and above 80°C, were not taken into account. In 2010, Vanin et al.[27] studied the combined effect of temperature and moisture content on dough viscosity for hydrothermal pathways related to bread crust, and future modeling studies of baking might benefit from these findings.

Oven-rise might be followed by a decrease in height or in perimeter of up to 30% from maximum height, with or without stabilization at prolonged baking times.[12,18] This phenomenon was widely reported for biscuits, cookies and short doughs[1,28] and was associated with the collapse of a non-rigid structure following the depletion of chemical leaveners and/or their enhanced percolation through the open porous structure. Sommier et al.[18] also suggested that cross-linking reactions were enhanced at high surface temperature by extensive protein coagulation and that Maillard browning reactions might be responsible for shrinkage. Simple drying at the surface may also provoke shrinkage; surprisingly this point has rarely been mentioned. To our knowledge, no modeling study has yet attempted to combine these different mechanisms of action to simulate the phenomenon of volume decrease occurring during baking and to evaluate their respective contributions.

Finally, porosity development during baking is far from homogeneous. There are locations where porosity increases and others where bubbles are squeezed.[19,29] This is affected by the time interval between the setting of the crust (imposing local inflation at constant overall volume) and the heating at the core. In other words, if the core is heated before the crust sets, little squeezing is observed.[19,20] However, this seems to be specific to low baking temperatures and not to apply to industrial baking conditions. Local porosity predicted by a baking model was recently compared with experimental porosity obtained dynamically by MRI (Magnetic Resonance Imaging),[2] with fairly good agreement. However, the numerical implementation of such a model could be improved. In particular, transport was considered unidirectional and cessation of overall expansion was not predicted by the model but imposed in the calculations, to fit the experimental observations to the dough scale. Most previous modeling studies have proceeded in the same way,[4,12,15] with the overall inflation distributed uniformly through the dough. The two-dimensional study carried out by Zhang et al.[5,6] demonstrated the feasibility of simulating porosity and overall deformation without imposing geometric parameters. Such a theoretical framework still requires validation by an experimental approach on a lower scale.

Table 1 compares the modeling approaches of three previous works: Zhang et al. 2005,[5] Nicolas et al. 2014,[3] and Lucas et al. 2015.[2] It can be observed from this table that many physical phenomena were taken into account in one of the three previous works but none of these studies included them all together. The present work takes into account all these phenomena (except the elastic part of product rheology) in a 2D geometry. In the modeling study proposed here, heat and mass transfer, including diffusion/conduction, convection modes, and evaporation-condensation were taken into account and developed as in previous studies.[2,3,5,6] Gaps from previous approaches were filled, as detailed below. CO_2 in both the liquid phase (dissolved) and in the gaseous phase was considered together with a temperature-dependent expression of production. Not only water vapor and CO_2 is considered in the gas phase but also air, this seems more realistic notably for the crust. The evaporation-condensation flux was introduced when pores were closed. Moreover, for the first time, the rheology is considered to depend both on temperature (direct influence and influence through starch gelatinization and gluten reticulation) and water content. Refining this input in a 2D implementation provided prediction of the cessation of overall expansion. Gravity and shrinkage due to water loss were also

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included to create the decrease in total volume during baking. The comparison of simulated porosity levels with those obtained experimentally and dynamically by MRI made it possible to discuss the mechanisms added to the model.

Materials and Methods

Sample preparation

Standard dough was prepared with 2000 g wheat flour (T55, 0.53% ash, 10.58% proteins, Moulins Soufflet Pantin, France), 40 g salt (La Baleine, France), 20 g improving baking powder (Freshbake improver, gluten formulation, B-24272, Puratos, Belgium), 100 g compressed yeast (l'Hirondelle, Lesaffre, France), and 1140 g water (uncontrolled quality). Flour was stored at −20°C and defrosted at ambient temperature 24 h before experiments. The moisture content of the flour was measured before each new experiment to check any changes and was maintained at 0.14 ± 0.005 kg kg^{-1} (wet basis). The ingredients were mixed (Moretti Forni, Spiry 8, Italy) at 100 rpm for 17 min. The final temperature of the dough was 25 ± 1°C. At the end of mixing, the dough was covered with plastic for 15 min, then divided into two and covered again with plastic for 10 min. Initial moisture content was determined from three dough samples after oven-drying at 104 ± 1°C for at least 24 h (estimated at about 0.453 ± 0.01 g (wet basis)). Sticks (length × diameter = 210 × 65 mm, ±1mm) were shaped manually from 200 g dough.

The shaped dough of approximately 200 g was placed in a glass net over which Teflon (Sansil LuborflonMS20, Argenteuil, France) had previously been spread to limit adherence. The edges of the dough sticks were thermally insulated with ceramic ensure two-dimensional (2D) heat transport. The net with the dough was then placed in a proving chamber (SFP12, BCR, France) with highly humidified air (>85%) at 35 ± 1°C, for approximately 45 min.

Experimental protocols and measurement methods

The dough was baked in a special convection oven that could be installed within the probe of an MRI device. The MRI oven is described full in previous works.[2,18] At t = 0, the mould was placed in the oven (air temperature 200 ± 1°C, mean air velocity 3 m s^{-1}).

Temperature and moisture content at different locations in the dough, CO_2 concentration in the oven and MRI images were measured continuously for 45 min, but not simultaneously (separate runs). Three runs (at least) were performed for each measurement, and variations between runs were represented in plots as error bars taking into account the value of one standard deviation ($\pm \sigma$).

Temperature was measured at different locations inside the dough (Figure 1), using optic fibers compatible with MRI measurements (FISO, 1 mm in diameter). Just before the proving step, fibers were inserted into the dough through a tiny hole at the bottom of the mould. The locations of the sensors were accurate to ±2 mm. In order not to disturb the baking process excessively, no more than two fibers were inserted at the same time and the core position (A) was recorded in all experiments to compare any given trial to any other. For crust measurement only, the optic fiber was deposited on the surface and could move with dough expansion.

Local moisture content was measured after interruption of baking at different times and sampling.

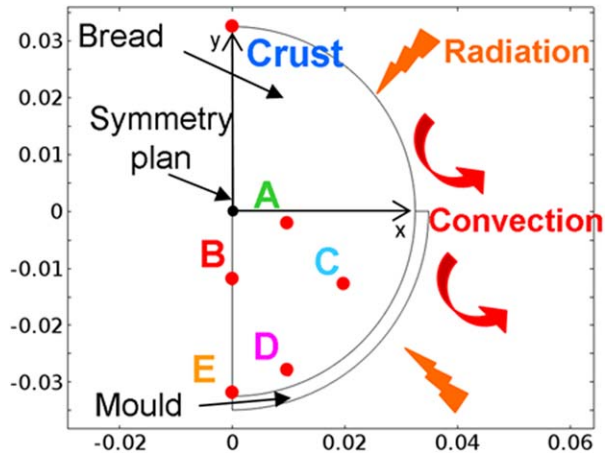


Figure 1. Symmetric bread and mould geometry with thermal exchanges; Positions of temperature evaluation points (Crust, A, B, C, D, E).

[Color figure can be viewed in the online issue, which is available at wileyonlinelibrary.com.]

CO_2 concentration in the oven was measured using Infra Red excitation and acoustic response analysis (Bruel & Kjaer, Denmark, type 1302) and corrected for air replacement in the oven to obtain the amount of CO_2 truly released from the loaf during baking. Further details can be found in Lucas et al.[30].

Local expansion of the dough was deduced from the relative displacement of oil microcapsules incorporated in the dough.[19,20] MRI contrast between oil and the dough made it possible to track the microcapsules; the densified crumb gave a signal as high as that of oil, and could limit the tracking of microcapsules in these regions. Interference of microcapsules with the baking process could be ignored since the surface of the microcapsules was small, their density was also close to that of the dough and their coating was resistant up to 120°C, thus avoiding oil release into the crumb during baking (T < 100°C). In contrast to the previously described experimental procedure,[19] sticks of agglomerated microcapsules were deposited on the laminated dough before manual rolling (shaping step).

MRI images were acquired using a SIEMENS OPEN 0.2T imager using the head antenna to receive the signal. MRI slices were positioned in the plane (x, y) of the receiver (transverse slice). Images were acquired with a Spin Echo (SE) sequence with the following parameters: TE (echo time) = 7.7 ms, TR (repetition time) = 420 ms, matrix (number of lines and columns) 128 × 128, FOV (field of view) (width and height of image) 80 × 128 mm^2, slice thickness 10 mm, 1 scan. The total time of one acquisition was 36 s; the mid-length of the sequence duration was taken into account when calculating the time to be associated with each image.

The dough section visible in the MRI image was divided into several triangles, using triplets of microcapsules. The surface of each triangle (S) was deduced from the microcapsule coordinates, and porosity from this surface was calculated as follows:

$$\varepsilon(t) = 1 - \frac{(1-\varepsilon(t_0))}{\frac{S(t)}{S(t_0)}}$$

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where porosity is expressed in $m^3_{(l+g)}$ $m^{-3}_{product}$, with $\varepsilon(t_0)$, the reference porosity just before baking, estimated at 0.90 (0.75 for gas fraction). For this estimation, a gas fraction of 10% after mixing and multiplication of the dough volume by 3.6 were applied, taking into account the two proving steps.

Physical Model

A multiphase model was written from the conservation equations of mass, energy and momentum.[31]

Mass conservation

The product to be modeled was composed of dough and bubbles, also called pores. It was assumed that the dough is mainly constituted of dry matter, liquid water, and CO_2 dissolved in water. Bubbles consist only of water vapor, CO_2, and air. They are characterized by the gas volume fraction relative to the product volume.

$$\frac{\partial \rho^a_{wl}}{\partial t}+\frac{\partial \rho^a_{wv}}{\partial t}+\vec{\nabla}\cdot\left(\vec{m}_{wl}+\vec{m}_{wv}+\left(\rho^a_{wl}+\rho^a_{wv}\right)\vec{u}\right)=0 \quad (1)$$

$$\frac{\partial \rho^a_{cl}}{\partial t}+\frac{\partial \rho^a_{cv}}{\partial t}+\vec{\nabla}\cdot\left(\vec{m}_{cl}+\vec{m}_{cv}+\left(\rho^a_{cl}+\rho^a_{cv}\right)\vec{u}\right)=I_c \quad (2)$$

With I_c the CO_2 gas generation by yeast activity.

$$\frac{\partial \rho^a_s}{\partial t}+\vec{\nabla}\cdot\left(\rho^a_s\vec{u}\right)=0 \quad (3)$$

$$\frac{\partial \rho^a_a}{\partial t}+\vec{\nabla}\cdot\left(\vec{m}_a+\rho^a_a\vec{u}\right)=0 \quad (4)$$

Water vapor and carbon dioxide vapor pressure were calculated by an equilibrium equation. Vapor pressure was calculated using the water activity and the saturation vapor pressure, which depends on temperature.

$$P_{wv}=a_w P_{wv;sat} \quad (5)$$

Henry's law expresses the link between molar carbon dioxide concentration in the liquid phase and carbon dioxide pressure in bubble gas.[32] Increase of water vapor and CO_2 pressure as well as temperature are taken into account by the ideal gas law to predict the porosity evolution.

$$P_{cv}=H_c\frac{X_{cl}/M_c}{X_{wl}/M_w} \quad (6)$$

Darcy's and Fick's laws are commonly used to model mass transfer by convection and molecular diffusion. At the beginning of baking, gas bubbles are considered unconnected. At this stage, Darcy's and Fick's fluxes are ignored in the gas phase. With temperature-dependent changes in molecular structures (starch, protein), the dough loses its extensibility, gas pressure increases and the film separating adjacent bubbles may rupture, making gas bubbles interconnected. In the model, a pore opening function (α) was used to simulate the increasing connectivity between bubbles as the baking proceeds (see Eqs. 7–9); for the reasons described above, it was only temperature dependent. A sigmoid function which uses values of zero before $T_{open}-\frac{\Delta T}{2}$ and 1 after $T_{open}+\frac{\Delta T}{2}$ was used ($T_{open}=62$°C and $\Delta T=10$°C). A sensitivity analysis regarding the parameters of the pore opening function was carried out. It showed that the pore opening temperature has an effect on global and local expansion (porosity) and carbon dioxide released. For example, if pore opening temperature is increased, global and local deformation increase and carbon dioxide release occurs latter. The value of T_{open} was chosen to be in accordance with experimental evolutions in terms of maximum height.

$$\vec{m}_{cv}=\alpha\left(-\omega_{cv}\frac{K_g}{v_g}\vec{\nabla}P-\rho^i_g D^{eff}\vec{\nabla}\omega_{cv}\right) \quad (7)$$

$$\vec{m}_a=\alpha\left(-\omega_a\frac{K_g}{v_g}\vec{\nabla}P-\rho^i_g D^{eff}\vec{\nabla}\omega_a\right) \quad (8)$$

Additional mass transfer occurs for water vapor transfer by evaporation-condensation. The evaporation-condensation flux (last term in Eq. 9) is proportional to the water vapor pressure gradient in the gaseous phase. Two types of resistance are involved, R_{bubble} for water vapor diffusion inside a bubble, R_{film} for the thermal conduction and the liquid water diffusion across the film between bubbles.

$$\vec{m}_{wv}=\alpha\left(-\omega_{wv}\frac{K_g}{v_g}\vec{\nabla}P-\rho^i_g D^{eff}\vec{\nabla}\omega_{wv}\right)-(1-\alpha)A_{wvvc}\vec{\nabla}P_{wv} \quad (9)$$

$$A_{wvvc}=\frac{\varepsilon_g}{R_{bubble}+R_{film}} \quad (10)$$

$$R_{bubble}=(1-\delta)\frac{RT}{M_w}\frac{(1-\omega_{wv})}{D_g} \quad (11)$$

$$R_{film}=\delta\left(\frac{1}{\rho^i_{wl}D_{wl}}\frac{\partial P_{wv}}{\partial X_{wl}}+\frac{L_w}{\lambda_{\mathrm{dgh}}}\frac{\partial P_{wv}}{\partial T}\right) \quad (12)$$

As explained before α corresponds to the pore opening phenomenon. When $\alpha=0$, pores are considered as unconnected and the second term of the equation is activated (evaporation condensation in gas bubble). When $\alpha=1$, the first term is activated with Darcy and Fick's law for water fluxes in the gas phase which can also lead to evapo-condensation. More details can be found in Lucas et al.[2].

Flux of Liquid water and dissolved carbon dioxide depends on their respective concentration gradients.[3,6]

$$\vec{m}_{wl}=-\rho^i_{wl}D_{wl}\vec{\nabla}X_{wl} \quad (13)$$

$$\vec{m}_{cl}=-\rho^i_{wl}D_{cl/wl}\vec{\nabla}X_{cl} \quad (14)$$

Energy conservation

In the dough, temperature is predicted by solving the enthalpy conservation equation.

$$\frac{\partial \hat{H}}{\partial t}+\vec{\nabla}\cdot\left(\vec{\dot{Q}}+\vec{H}+\hat{H}\vec{u}\right)=0 \quad (15)$$

$$\text{where } \hat{H}=\rho c_p\left(T-T_{ref}\right)+\rho^a_{wv}L_w+\rho^a_{cv}L_c \text{ is the volume enthalpy of the product;} \quad (16)$$

with ρ and c_p product density and heat capacity, respectively, and T_{ref} the reference temperature at 0°C.

$$\vec{\dot{Q}}=-\lambda\vec{\nabla}T \text{ is the conduction heat flux} \quad (17)$$

$$\vec{H}=\vec{m}_{wl}c_{p;wl}\left(T-T_{ref}\right)+\vec{m}_{wv}\left(c_{p;wv}\left(T-T_{ref}\right)+L_w\right)+\vec{m}_{cl}c_{p;cl}\left(T-T_{ref}\right)+\vec{m}_{cv}\left(c_{p;cv}\left(T-T_{ref}\right)+L_c\right)+\vec{m}_a c_{p;a}\left(T-T_{ref}\right) \quad (18)$$

is the enthalpy transfer relative to mass transfer.

In the mould there is only conduction with constant conductivity.

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$$\rho_{md}c_{p;md}\frac{\partial T_{md}}{\partial t}+\vec{\nabla}\cdot\left(-\lambda_{md}\vec{\nabla}T_{md}\right)=0 \tag{19}$$

Momentum conservation

Momentum conservation is used to predict dough overall deformation and porosity. Dough is considered as a viscous fluid with internal gas pressure stress. This viscous model is only an approximation of the real very complex rheological behavior of bread dough at different baking stages. The viscous approximation is rather good before starch gelatinization and dehydration. After gelatinization and dehydration the product becomes rigid, this is taken into account here by a high value of viscosity which ensures low deformation. This approximation remains then acceptable for the prediction of porosity evolution (see "Results and Discussion") even if it does not reflect the real rheological behavior.

$$\frac{\partial(\rho\vec{u})}{\partial t}+\vec{\nabla}\cdot\left(\overline{\overline{\tau}}+(P-P_{atm})\overline{\overline{I}}+\rho\vec{u}\vec{u}+\overline{\overline{\sigma}}_{sh}\right)=\rho\vec{g} \tag{20}$$

where $\overline{\overline{\tau}}=\eta(-(\vec{\nabla}\vec{u}+\vec{\nabla}\vec{u}^{t})+\frac{2}{3}(\vec{\nabla}\cdot\vec{u})\overline{\overline{I}})$ is the viscous stress tensor and $P=P_{wv}+P_{cv}+P_a$ the gas pressure.

The expression of $\overline{\overline{\sigma}}_{sh}$ (Eq. 24) considers that shrinkage occurs due to water evaporation (decrease in liquid moisture content). The underlying assumption is that the volume of water lost by evaporation is not replaced by a gaseous phase but is lost locally by the product itself.

The rate of water loss, expressed in terms of volume of water by time unit and volume unit of product is:

$$\frac{\partial V_{water}}{\partial t}\Big/V_{product}=\varepsilon_s\frac{\rho_s^i}{\rho_{wl}^i}\frac{\partial X_{wl}}{\partial t} \tag{21}$$

It is assumed that this relative volume change induces no additional stress and an isotropic deformation whose velocity divergence is equal to:

$$\vec{\nabla}.\vec{u}=\varepsilon_s\frac{\rho_s^i}{\rho_{wl}^i}\frac{\partial X_{wl}}{\partial t} \tag{22}$$

This results (in 2D) in a viscous stress of:

$$\overline{\overline{\tau}}=\eta\left(-\left(\vec{\nabla}\vec{u}+\vec{\nabla}\vec{u}^{t}\right)+\frac{2}{3}\left(\vec{\nabla}\cdot\vec{u}\right)\overline{\overline{I}}\right)=-\frac{1}{3}\eta\left(\vec{\nabla}\cdot\vec{u}\right)\overline{\overline{I}} \tag{23}$$

which is compensated by the additional term $\overline{\overline{\sigma}}_{sh}$. This lead to following expression

$$\overline{\overline{\sigma}}_{sh}=\frac{1}{3}\eta\,\varepsilon_s\frac{\rho_s^i}{\rho_{wl}^i}\frac{\partial X_{wl}}{\partial t}\overline{\overline{I}} \tag{24}$$

Boundary conditions

Boundary conditions vary according whether the product is in contact with the mould or the product is in contact with the oven atmosphere. A symmetry condition (no flux) is applied at the plane of symmetry (Figure 1).

Surface of the Product in Contact with Oven Atmosphere. The mass flux due to convection in the boundary layer at the interface between dough and the surrounding atmosphere depends on the difference between the gas concentration (j = wv for water vapor, j = cv for CO_2, j = a for air) at the dough surface and in the oven.[32]

$$\dot{m}_{j,b}=k_{conv}\frac{M_j}{R}\left(\frac{P_j}{T}-\frac{P_{j,oven}}{T_{oven}}\right) \tag{25}$$

The convection mass transfer coefficient is calculated using the Chilton-Colburn analogy.[31]

$$k_{conv}=\frac{h_{conv}}{\rho_{a,oven}c_{p;a,oven}Le^{2/3}} \tag{26}$$

In addition, there is Darcian flow at the surface which is proportional to the pressure difference between the surface of the product and the oven.

$$\dot{m}_{db}=\xi(P-P_{oven}) \tag{27}$$

As the flow resistance is very low outside the product a high value of ξ was taken so that $P-P_{oven}\ll P_{oven}=P_{atm}$. If $P>P_{oven}$

$$\dot{m}_{j,db}=\omega_j\dot{m}_{db} \tag{28}$$

or if $P<P_{oven}$

$$\dot{m}_{j,db}=\omega_{j,oven}\dot{m}_{db} \tag{29}$$

Displacement velocity equation

No constraint is applied at the surface.

$$-\vec{n}\cdot\left(\overline{\overline{\tau}}+(P-P_{atm})\overline{\overline{I}}+\rho\vec{u}\vec{u}+\overline{\overline{\sigma}}_{sh}\right)=0 \tag{30}$$

Heat equation

Convection heat flux is applied at the product surface (Figure 1). The overall heat transfer coefficient h_{surf} takes into account both convection and radiation heat transfer $h_{surf}=h_{conv}+h_{rad}$. To this transfer must be added energy exchanges associated with mass flux.

$$\begin{aligned}-\vec{n}\cdot\left(\dot{Q}+\dot{H}\right)&=h_{surf}(T_{oven}-T)\\&-\left(\dot{m}_{wv,b}+\dot{m}_{wv,db}\right)\left(c_{p;wv}\left(T-T_{ref}\right)+L_w\right)\\&-\left(\dot{m}_{cv,b}+\dot{m}_{cv,db}\right)\left(c_{p;cv}\left(T-T_{ref}\right)+L_c\right)\\&-\left(\dot{m}_{a,b}+\dot{m}_{a,db}\right)c_{p;a}\left(T-T_{ref}\right)\end{aligned} \tag{31}$$

Surface of the Mould. At the mould/air interface (Figure 1), convection and radiation heat flux is applied as for the product/oven atmosphere interface. At the mould/product interface, the heat flux is controlled by a contact exchange coefficient h_{cont}; there is no mass flux and no displacement of matter.

Properties

Model parameters are presented in this section and in the Table 2.

Water activity

The water activity expresses the relationship between the moisture content and vapor pressure.[33]

$$a_w=\frac{P_{wv}}{P_{wv;sat}}=\exp\left(\frac{-8}{(100X_{wl})^{1.3}}\right) \tag{32}$$

Henry's law

The constant of Henry's law H_c depends on temperature.[32]

$$H_c=\frac{1}{2.88\,10^{-6}\exp\left(-2.07\,10^{-2}T\right)} \tag{33}$$

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Table 2. Input Parameters Used in the Model

Symbol	Definition	Units	Value
$C_{p;wl}$	Specific heat of liquid water[32]	J kg^{-1} K^{-1}	4180
λ_{wl}	Thermal conductivity of liquid water[32]	W m^{-1} K^{-1}	0.68
λ_a,λ_{cv},λ_{wv}	Thermal conductivity of air, CO_2, and water vapor[32]	W m^{-1} K^{-1}	0.024, 0.025, 0.02
h_{surf}	Global heat transfer coefficient at top of loaf	W m^{-2} K^{-1}	$h_{surf}=h_{conv}+h_{rad}$ with $\begin{cases} h_{rad}=4\sigma\left(\frac{T_{oven}+T_0}{2}\right)^3=10 \\ h_{conv}=10 \end{cases}$
h_{cont}	Heat transfer coefficient at the mould/product interface (estimated)	W m^{-2} K^{-1}	100
T_{oven}	Air temperature in oven	°C	200
ρ^i_{wl}	Density of liquid water[32]	kg m^{-3}	1000
D_g	Binary diffusivity of water vapor in CO_2[32]	m^2 s^{-1}	0.26 10^{-4}
g	Gravitational acceleration	m s^{-2}	9.81
M_a, M_c, M_w	Molar mass of air, CO_2 and water [32]	kg mol^{-1}	0.029, 0.044, 0.018
$P_{wv.sat}$	Saturated water vapor pressure	Pa	$P_{v.sat}=101325\times 10^{\left(17.443-\frac{2795}{T}-3.868Log_{10}T\right)}$
P_{oven}	Total gas pressure in oven	Pa	Atmospheric pressure
$P_{wv,oven}$	Partial water vapor pressure in the oven	Pa	1 412
$P_{cv,oven}$	Partial carbon dioxide vapor pressure in the oven	Pa	10
$P_{a,oven}$	Partial air pressure in the oven	Pa	$P_{oven}-P_{wv,oven}-P_{cv,oven}$
ξ	Surface Darcian flow coefficient	s m^{-1}	10^{-4}
δ	Ratio of wall thickness to total wall and bubble thickness	m m^{-1}	$\delta=1-\varepsilon_g^{1/3}$
v_g	Kinematic viscosity of gas[32]	m^2 s^{-1}	3 10^{-5}
τ	Tortuosity of liquid phase due to presence of bubbles (calculated for a sphere)	–	$\pi/2$

Liquid water and dissolved carbon dioxide diffusivity

Diffusivity of liquid water depends on moisture content.[6]

$$\text{with, } D_{wl}=10^{-9}\exp\left((-2.8+2X_{wl})\varepsilon\right) \tag{34}$$

The diffusivity of dissolved carbon dioxide is assumed to be equal to the carbon dioxide diffusivity in liquid water weighted by the liquid fraction annotated ε_l and tortuosity annotated τ.

$$D_{cl/wl}=2.8\ 10^{-9}\frac{\varepsilon_l}{\tau} \tag{35}$$

Gas permeability

Permeability[34] takes the value of 1×10^{-11} m^2.

Effective diffusivity of gas

The effective diffusivity of gas depends on the porosity, liquid saturation, and gas diffusivity in a free medium.[6]

$$D^{eff}=D_g\left((1-1.11S)\varepsilon\right)^{4/3} \tag{36}$$

Generation of Carbon Dioxide. Generation of carbon dioxide is temperature dependent. A constant value of 0.347 10^{-4} kg CO_2 per m^3 of a product and unit of time (s) is assumed below 40°C.[24,35] This rate was deduced from that observed experimentally during proving (constant-rate period) under the conditions (yeast type, temperature) used for the experimental validation of the present model. As yeast activity is absent above 50°C, it is assumed that CO_2 generation decreases linearly from 40 to 50°C.

Intrinsic densities and specific heat

The intrinsic density[36] of dry matter is equal to 1800 kg m^{-3} and specific heat[36] to 1800 J kg^{-1} K^{-1}. Ideal gas law is applied to evaluate gas density. Gas-specific heat is equal to 1000 J kg^{-1} K^{-1} for air, 871 J kg^{-1} K^{-1} for carbon dioxide and 1842 for water vapor. The latent heat of vaporization is equal to 2300 10^3 J kg^{-1} for water (L_w) and 573 10^3 J kg^{-1} for carbon dioxide (L_c).

Thermal conductivity

The Maxwell conductivity model was used.[37] The continuous and dispersed phases used in this model were the dough without gas (dgh) and gaseous (g) phases, respectively.

$$\lambda=\lambda_{\text{dgh}}\left(\frac{2\lambda_{\text{dgh}}+\lambda_g-2\varepsilon_g\left(\lambda_{\text{dgh}}-\lambda_g\right)}{2\lambda_{\text{dgh}}+\lambda_g-\varepsilon_g\left(\lambda_{\text{dgh}}-\lambda_g\right)}\right) \tag{37}$$

with

$$\lambda_{\text{dgh}}=\lambda_s\frac{\varepsilon_s}{\varepsilon_s+\varepsilon_l}+\lambda_l\frac{\varepsilon_l}{\varepsilon_s+\varepsilon_l} \tag{38}$$

$$\lambda_g=\omega_{cv}\lambda_{cv}+\omega_a\lambda_a+\omega_{wv}\lambda_{wv} \tag{39}$$

The intrinsic solid thermal conductivity was temperature dependent[37]:

$$\lambda_s=0.20141+1.3874\ 10^{-3}(\text{T}-273.15)+4.3312\ 10^{-6}(\text{T}-273.15)^2 \tag{40}$$

Dynamic viscosity of bread dough

Previous models have used constant[1] or temperature-dependent[3,6] dynamic viscosity. In a previous study,[27] dynamic viscosity was measured with a DMTA rheometer at

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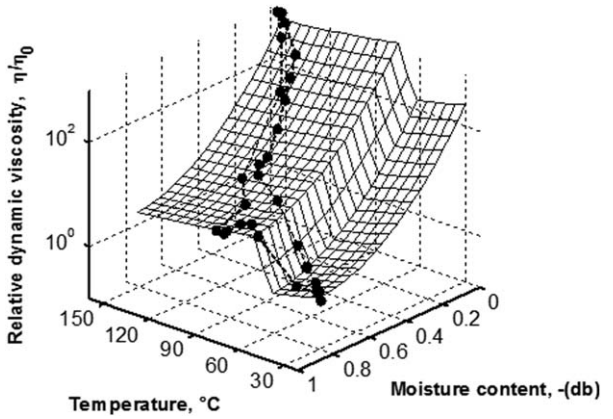


Figure 2. Modeled (mesh) relative dynamic viscosity with moisture content and temperature; experimental measurement of relative dynamic viscosity (black circles).

different moisture content and temperature conditions. The dynamic viscosity model used was fitted with the experimental data. It depends on temperature, baking degree, and moisture content (Figure 2). Different pathways were followed during these rheological measurements corresponding to the evolution of crumb (maintaining high water content during temperature increase) for some samples and for crust (dehydrating during temperature increase) for other ones. Thus, it can be assumed that the model also applies for intermediate cases. Baking degree was assumed to depend on the maximum temperature reached by the product (T_{max}). When considering only the baking stage: $T_{max} = T$. However this is not the case if cooling is included in the simulation.

Many phenomena are taken into account in the viscosity model through the influence of temperature and moisture content.

$$\eta = \eta_{ref} 10^{(G_T + G_{Xwl})} \tag{41}$$

First, viscosity decreases as for water (without transformation) when temperature increases between 20 and 60°C. This effect is modeled by the first part of Eq. 42. Then, starch gelatinization is taken into account with an increase of viscosity (with a factor of about 10) between 60 and 70°C using a sigmoid function $\beta(T)$ which takes a value of zero before 60°C and the unity after 70°C. This is taken into account in the second part of Eq. 42.

$$G_T = 247.8\left(\frac{1}{T-140} - \frac{1}{T_{ref}-140}\right) + p_3\left(\beta(T_{\max}) - \beta(T_{ref})\right) \tag{42}$$

The last phenomenon included is the dehydration effect (crust) which increases the viscosity (with a factor of about 100). This is taken into account in Eq. 43.

$$G_{Xwl} = -p_1\left(X_{wl} - X_{wl;ref}\right) + p_2\left(X_{wl} - X_{wl;ref}\right)^2 \tag{43}$$

With constant parameters: $p_1 = 1.031$; $p_2 = 2.499$; $p_3 = 0.936$; $\eta_{ref} = 5\ 10^4$Pa.s; $T_{ref} = 30.5$°C; $X_{wl;ref} = 0.83$ kg kg_{dm}^{-1}.

Mould properties

The oven was placed in a MRI device during baking to evaluate the geometric evolution of the bread. Because of the magnetic field, the mould was composed of Pyrex glass. The thermal properties of the mould were as follows.[32]

$$\rho_{md} = 2225 \text{ kg m}^{-3};\ c_{p;md} = 835 \text{ J kg}^{-1}\text{K}^{-1};$$
$$\lambda_{md} = 1.4 \text{ W m}^{-1}\text{K}^{-1}$$

Initial Conditions

Experimental initial conditions were used for temperature and moisture content.

$$T_0 = 30.5°\text{C};\ X_{wl0} = 0.83 \text{ kg/kg}_{dm}$$

The volume solid fraction was evaluated by using the initial gas fraction (0.75 m_g^3/m_{tot}^3, $\varepsilon_{s0} = 0.10\ m_s^3/m_{tot}^3$).

Gas bubbles were assumed to be saturated with carbon dioxide gas, corresponding to the following initial dissolved carbon dioxide content:

$$X_{cl0} = 2.35\ 10^{-3} \text{kg/kg}_{dm}$$

For gas pressure, the weight of the solid matrix was taken into account. The initial pressure was space dependent.

$$P_0 = \rho g\left(\sqrt{r_0^2 - x_0^2} - y_0\right) + P_{atm} \tag{44}$$

where r_0, x_0, and y_0 are, respectively, the initial radius and the position (x,y) at initial time.

Convection heat transfer was evaluated at 10 W m^{-2} K^{-1}. Air humidity was considered to be 60% at a temperature of 25°C, and the equivalent vapor pressure in the oven was equal to 1411 Pa. Carbon dioxide partial pressure is very low in the atmosphere (about 0.0037% molar fraction). A value of 35 Pa was imposed. With an atmospheric pressure of 101,325 Pa, the air partial pressure (nitrogen and oxygen) is equal to 99, 904Pa.

Numerical Solution

From the conservation equations, a system of partial derivative equations was determined involving six state variables; liquid moisture content X_{wl}, dissolved carbon dioxide content X_{cl}, gas pressure P, solid volume fraction ε_s, temperature T, and displacement velocity $\vec{u}$ (See Appendix). The model was solved with the Comsol Multiphysic v4.4 commercial software. The modeled geometry was a half disk with an initial diameter of 6.5 cm to take into account the plane of symmetry.

A mesh sensitivity study was performed to optimize the number of elements. This study compared meshes where the maximum element size was, respectively, 3, 2, 1.5, and 1 mm and showed no significant differences in term of temperature and porosity evolution when using more elements smaller than 2 mm. The corresponding mesh was composed of 4369 triangular elements and was refined at the surface (Supporting Information 1). An ALE (Arbitrary Lagrangian-Eulerian method) moving mesh was used. A quadratic element order was applied, except for the momentum equation which was cubic. Relative tolerance was imposed at 10^{-4}. A CPU time of 22 h was needed for simulations of 45 min.

Results

The numerical results are presented first and compared to experimental data of different types. Finally, overall

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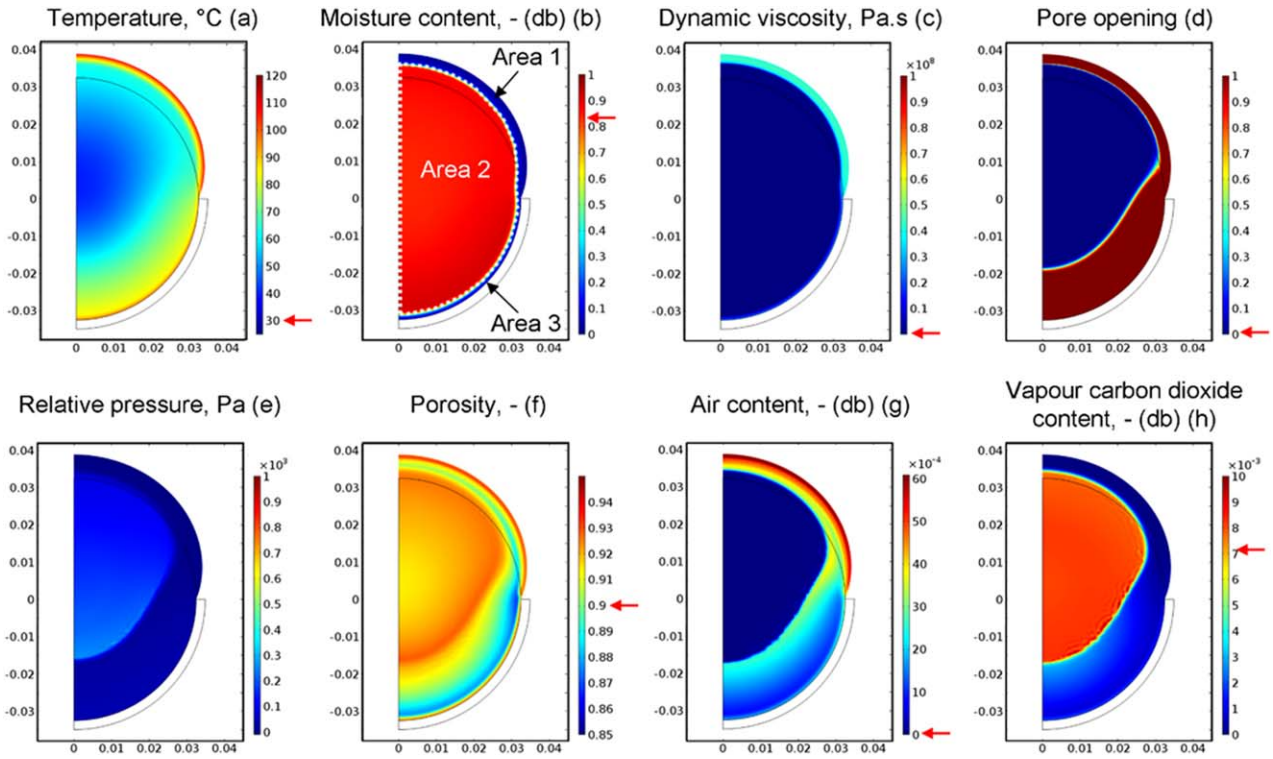


Figure 3. Simulation maps at 7 min of baking for (a) temperature, (b) moisture content, (c) dynamic viscosity of dough/crumb, (d) pore opening function, (e) relative gas pressure, (f) porosity, (g) air content, and (h) carbon dioxide vapor content.

The black line shows the initial geometry of the bread and mould. The red arrow near each color bar indicates the initial value. [Color figure can be viewed in the online issue, which is available at wileyonlinelibrary.com.]

deformations and porosity are presented to improve understanding of the effects of mechanisms (gravity, carbon dioxide generation, and shrinkage due to dehydration) on the simulated deformation. Note that the impact of these mechanisms on temperature and mass evolution was insignificant.

Since transport and deformation were a strongly two directional, maps of different input parameters or variables are presented in Figure 3, and will serve in the discussion of the following results. For illustrative purposes, 7 min of baking was selected for Figure 3; other maps at different baking times (2 min 30 s and 15 min) are presented in Supporting Information 3 and 4.

Temperature

Simulated temperatures were in relatively good agreement with experimental temperatures (Figure 4). Far from the dough surface (A, B), the temperature increased slowly for the first 5 min. Between 5 and 15 min, the temperature increase accelerated, explained by the evaporation-condensation phenomenon which transferred warm vapor toward the centre of the bread; this was indicated by the increase in moisture content in the crumb within this time period (Figures 6b, c), a point that will be discussed later. After 15 min, the temperature stayed constant at around 100°C. This was the boiling temperature of water expected in a product saturated with vapor at atmospheric pressure. Indeed, water activity at these points remained close to unity. The pressure inside the bubbles was only about 100 Pa above the atmospheric pressure (see Supporting Information 5 for baking times higher than 15 min). The presence of a small amount of CO_2 and air in the bubbles showed that the temperature was slightly lower than 100°C. The same evolution was observed at the intermediate position between the core and the surface (e.g., C), with a shorter initial period of low temperature increase. Note that the coldest point in the simulation was not located at the barycentre of the dough section but was instead in the top part, at about two thirds of the total height (Figure 3a); this result could not be checked experimentally since the fibers were positioned before proving to avoid local or even total collapse of dough, resulting in instrumentation only in the lower part of the dough after proving (see "Materials and Methods" section).

As expected at the top surface (crust), the temperature increased faster than anywhere else, and exceeded the boiling temperature of pure water at atmospheric pressure. This is explained by the lowering in moisture content of these layers by drying and thus of water activity. The evolution was the same for temperatures near the mould (D, E), with the same explanation as for the top surface (discussed further in the next section).

Optimal baking time can be defined as a brown-colored crust (not too dark) and a core crumb presenting the dough-crumb transition, if possible with starch gelatinization as complete as possible.[2] These two events can be evaluated from time-temperature curves inside the dough with ideal coloration of the crust corresponding to approximately 140°C[38] and fully swollen starch granules by applying a holding time of 10 min at temperatures above 95°C.[39] These two thermal conditions were encountered at about 20–25 min, and the process was prolonged beyond this point for the initial purpose of further validating the drying phenomenon.

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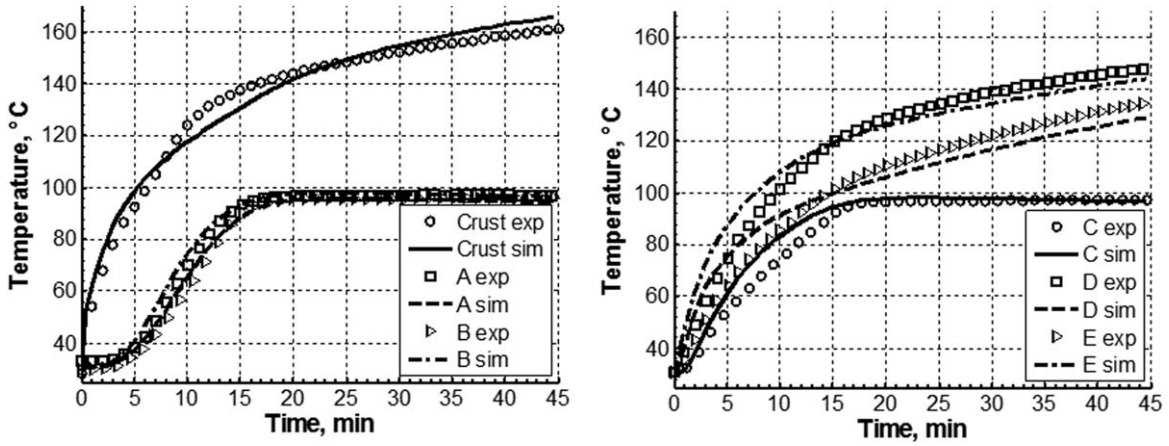


Figure 4. Experimental and simulated temperatures at different locations in the dough (A, B, C, D, E) (presented in Figure 1).

Total mass of water and moisture content in different regions

Time course changes in simulated water mass loss were in agreement with those observed experimentally (Figure 5). Water mass decreased continuously and almost linearly with time; this was consistent with previous reports in the literature.[2,3,12,40] The loss represented 20–25% and 45% of the initial amount of water at the optimal baking time and after 45 min of baking, respectively. Two periods were observed for the evolution of water mass loss. The drying rate during the first 17 min was lower than for the end of baking. The transition between the two periods corresponded to the cessation of the evaporation-condensation phenomenon at the core.[2]

Three areas of local moisture content were found, in both the simulation and in the experiment: a dry zone at the top of the dough (Figure 6, also area 1 in Figure 3b), a large zone of high moisture content in the centre of the dough (Figure 6, also area 2 in Figure 3b) and another a dry zone at the bottom, in the proximity of the mould (Figure 6, also area 3 in Figure 3b). The top dry zone was the consequence of water evaporation at the surface in contact with the dry atmosphere of the oven. The heat pipe phenomenon (evaporation-condensation) also contributed to drying the surfaces.[2] In the model, this transfer was driven by the water vapor pressure gradient which mainly depended on the temperature gradient. Since the temperature was lower at the centre of the dough at the beginning of the baking process, vapor was transferred from the dough surface to the centre. This explains why there was also a dry zone that developed in the proximity of the mould. Both this mechanism and its set of equations were validated experimentally by a decrease in the moisture content at the bottom surface and an increase in moisture content at the centre of the bread.

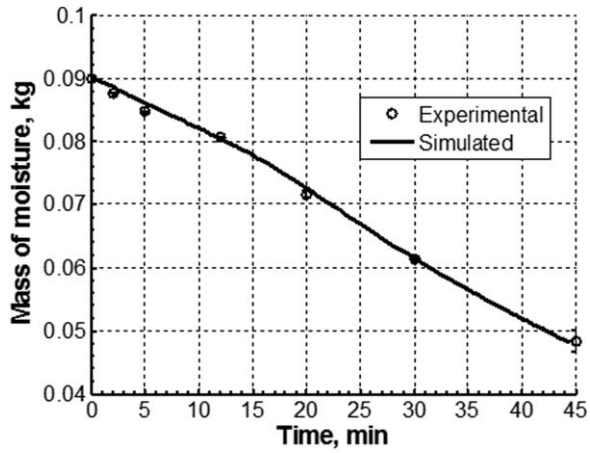


Figure 5. Simulated (solid line) and experimental (circle) mass of moisture in the dough during baking; the experimental standard deviation is presented as error bars but they are hardly visible since its level was low (±0.0005 kg).

However some differences occurred between simulation and experiment. In the simulation, moisture content in the bottom crumb (to the exclusion of the bottom crust) was slightly higher than in the top crumb (Figure 6c for instance). Such a trend was not observed in the experiment, probably because the adhesion of the product to the mould was far from perfect, allowing water vapor to escape along the internal surface of the mould. Moreover, experimental moisture content levels in the crust were higher than simulated levels, explained by a rehydration process occurring in the experiment during the cooling step that preceded sampling which was not simulated in the results presented in Figure 6. Additional simulations of the baking process were carried out followed by a cooling step. They showed variations in moisture content of up to $+0.08$ kg kg_{dm}^{-1} in the peripheral layers of the dough and -0.1 kg kg_{dm}^{-1} at the core for 10 min of cooling (Supporting Information 2).

Carbon dioxide released by the loaf

The total amount of carbon dioxide released in the oven obtained by simulation was in relatively good agreement with the experimental value (Figure 7a). However, a discrepancy in the slope was noticeable because the simulation was faster than the experiment.

The carbon dioxide mass balance showed that, in the case of fermented bread dough, the carbon dioxide was mainly in the gaseous phase at the beginning of baking. Dissolved carbon dioxide represented 12.44% of the total amount, and carbon dioxide generated by the activity of yeasts during baking represented an even smaller part of the total carbon dioxide released in the oven (4.8%, Figure 7b). Yeast activity was shown to have a limited effect (Figure 7b). Since there is controversy in the literature about the temperature at which yeasts can survive, higher upper limits for CO_2 generated by yeast

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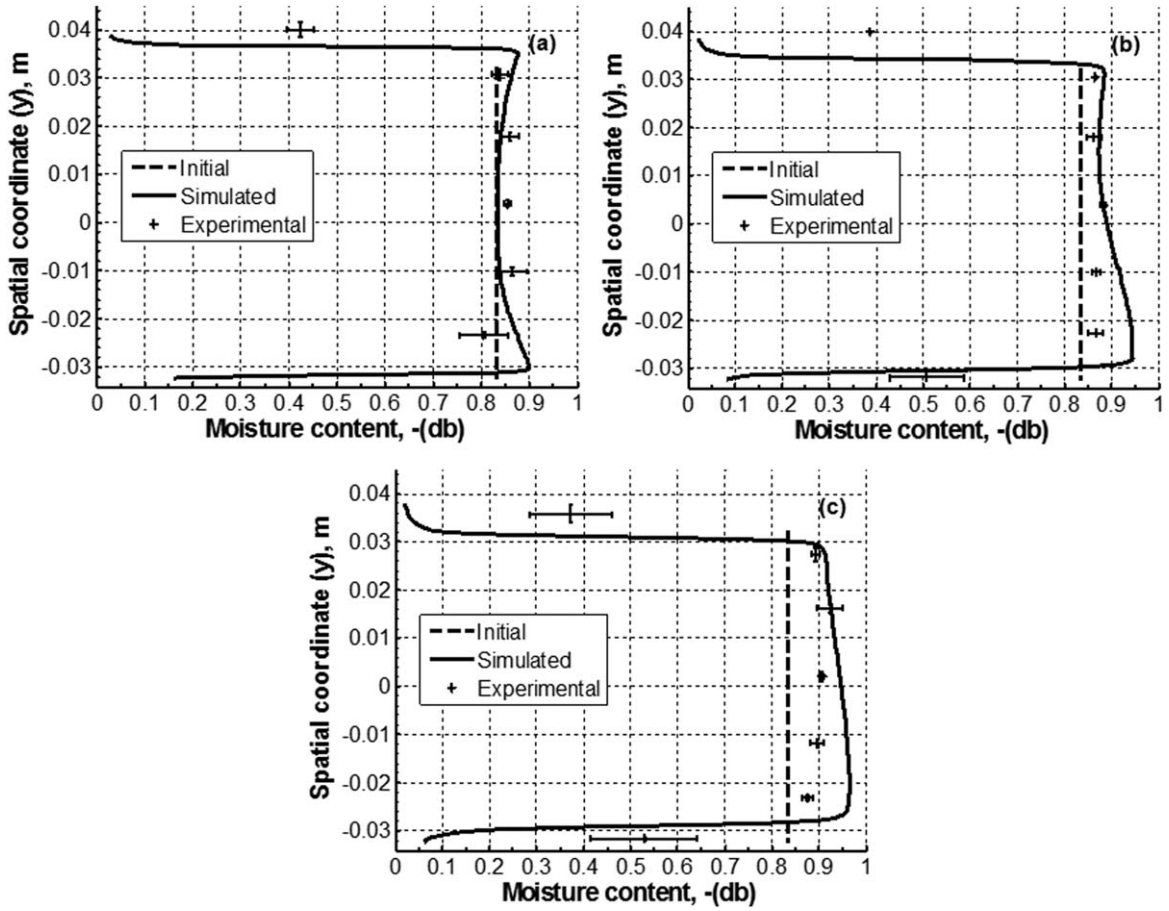


Figure 6. Simulated and experimental moisture content in dough sampled along the line of symmetry at 5 (a), 12 (b) and 20 min (c) of baking; the dotted line represents the initial value.

(up to 70°C) were also tested, but the results were not different.

At 15 min of baking, most of the CO_2 was released into the oven atmosphere (Supporting Information 4, t = 15 min). When carbon dioxide decreased locally, the gas pressure tended to the atmospheric pressure and was replaced by air or water vapor by convection and diffusion mass transfer. This occurred in already open pores, the highest air content being observed where the moisture content was the lowest, as in the top crust (Figure 3g).

Overall deformation and local porosity evolution

Comparison Between Simulation and Experiment Results. Simulated relative height evolution was in very good agreement with the experimental evolution (Figure 8a). Rapid expansion (near 11%) was observed for 5 min, reaching

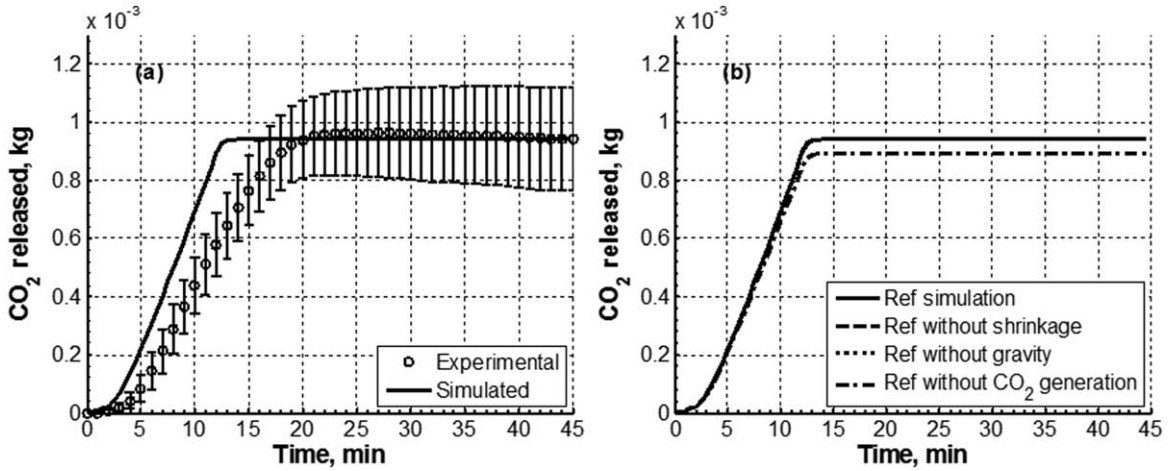


Figure 7. Simulated (solid line) and experimental (circle) carbon dioxide released by the loaf (a); study on the simulated carbon dioxide released by the loaf (b).

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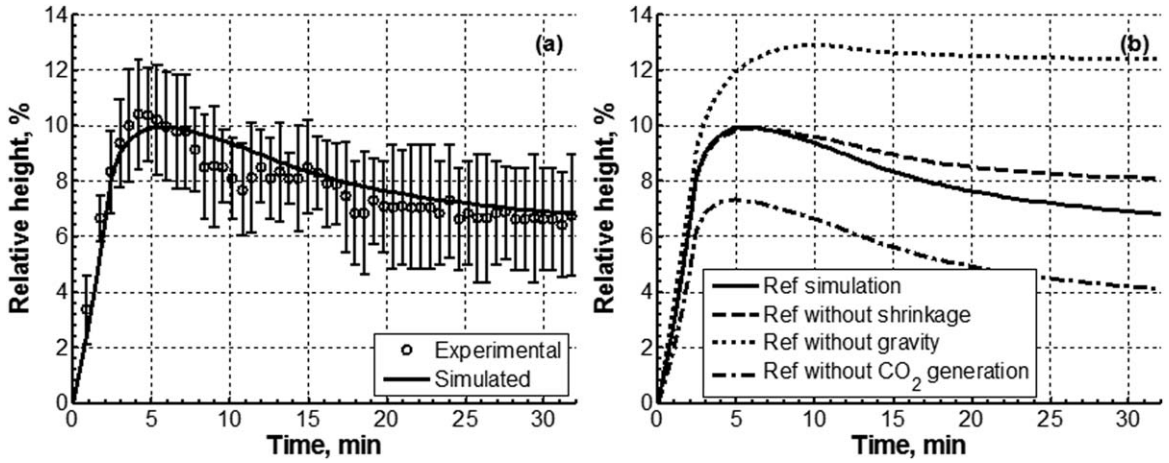


Figure 8. Simulated (solid line) and experimental (circle) total height of dough relative to its initial value (a); study on the simulated relative height during baking (b).

maximum expansion of 11% (relative to the initial total height of the dough) and followed by a continuous decrease (gain in total height at 30 min reduced to 7% of the initial total height).

The model was also able to reproduce the genuine shape of the bread section, with dough bulging near the top of the mould (Supporting Information 3 and 4, t = 15 min).

Experimental and simulated local porosity was also compared at different locations in the dough section (Figure 9, also Figure 3f). The model was able to reproduce the general trends observed experimentally, namely monotonic inflation followed by stabilization in the core and top crumb, and successive inflation and squeezing in other regions. Some differences in rate (rate of inflation or of squeezing) between simulation and experimental conditions were noticeable in several regions. However, transition times (between inflation and stabilization or squeezing) were between the simulation and the experiment, and the averaged porosity values observed experimentally at the end of baking were reproduced satisfactorily by the model for all regions except the top crust where the decrease in porosity was also not reproducible experimentally (the simulation was within the confidence interval of the experiment). It should be noted that the porosity at the end of baking differed between regions. Expansion of the top and core crumb was greater than the bottom crumb (3.3% compared to 1.1% in the bottom crumb at the same baking time). Porosity values in the top and bottom crust regions were well below their initial values (end of the proving step). Levels of inflation in the very top and very bottom crust were high from the beginning of baking (Figure 3f) and as high as that in the core taken after 15 min of baking (Supporting Information 4, t = 15 min); this was formerly related to the absence of squeezing due to high viscosity.[2] These regions were not accessible by MRI (a higher spatial resolution would be required) and thus they could not be validated in this study.

Detailed Analysis of Local Expansion. All the trends presented above are discussed below in the light of complementary analysis of simulated data. Figures 10 and 11 present the porosity profiles at different baking times, together with the effects of different mechanisms possibly involved in the inflation or compression occurring locally. Figure 10 plots pressure profiles in relation to viscosity and pore opening; gravity stress, representing the weight of product located above a given vertical position (y), was also added to Figure 10 and compared to the pressure (net balance). A complementary approach based on the suppression of shrinkage due to dehydration, gravity or CO_2 generation is plotted in Figure 11; according to the resulting porosity profiles.

Baking times were chosen according to the progress of pore opening. The pores began to be connected in the outer dough layers of the product at about 2 min 30 s, and were connected everywhere after 11 min; baking times of 2 min 30 s, 7 min, and 15 min were chosen.

Figure 11 presents the porosity profiles along the vertical line of symmetry for the reference simulation and when one mechanism (shrinkage, gravity, or CO_2 generation) is omitted.

Local Inflation in the Product at Early Baking Times. Dough expansion can be explained by the following phenomena, either directly or indirectly dependent on temperature: thermal dilatation of gases, particularly those initially present at the beginning of baking in gas bubbles, water vaporization, carbon dioxide dissolved in the liquid phase and released in the gaseous phase, and carbon dioxide generation by yeast, which is also released in the gaseous phase. With the rise in temperature, gas pressure in closed pores will increase according to the ideal gas law. Low viscosity of the dough walls separating the bubbles and low gravity stress compared to pressure are prerequisites for bubble inflation. When the dough was placed in the oven, heat transfer occurred in the product which increased the temperature at outer layers first, and the progressed toward the core. This explains why porosity increased mainly near the top and bottom regions during the first 2 min 30 s (1 cm thick in Figure 11a). The associated pressure level (Supporting Information 5) was not measured in this study, but was consistent with previous reports in the literature.[3,41] Consistent with this inflation, Figure 10 shows that at the beginning of baking (2 min 30 s) almost all pores were closed (Figure 10c1) and the gas pressure everywhere was higher than the gravity stress (Figure 10b1).

Despite the small amount of carbon dioxide generated (see "Carbon dioxide released by the loaf" section), CO_2 generation had a significant impact on inflation, both overall and locally. Without CO_2 generation, the relative total height of the bread was significantly lower (−42%, Figure 8b) at the end of baking. In fact, carbon dioxide generation permitted inflation at the centre early in baking (Figure 11a) while the bread temperature remained close to that of the proving

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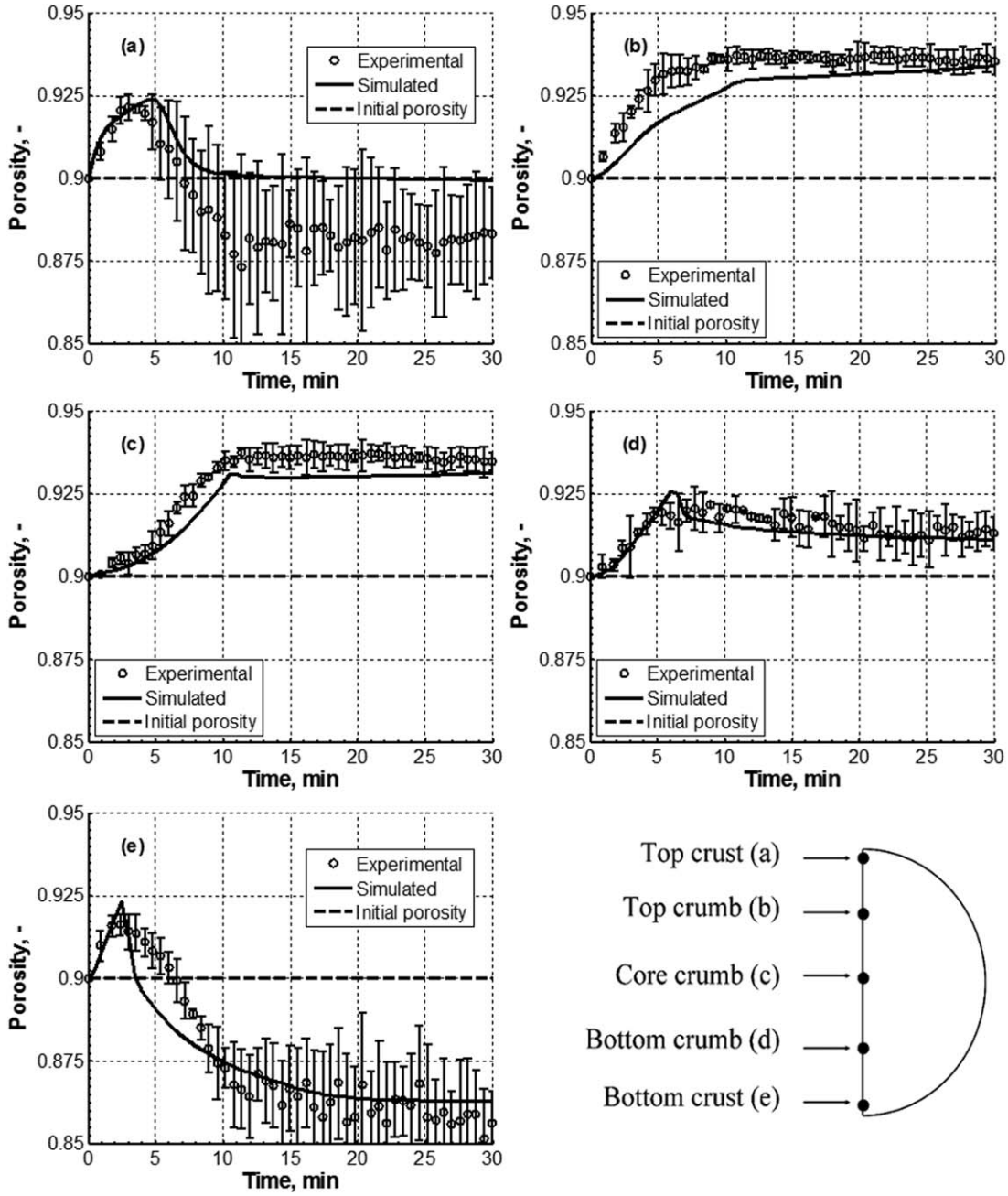


Figure 9. Simulated (solid line) and experimental (circle) porosity during baking represented for different locations along the symmetry line.

The dotted line represents the initial value.

chamber (its initial value). The observed difference in porosity with and without CO_2 generation was then preserved at longer baking times (t = 7–15 min Figures 11b, c).

At longer baking times, pores opened and the crust was less rapidly deformable, making the interpretation of porosity changes more complicated. Transition occurred between 2 min 30 s and 5 min. At 2 min 30 s, the increase in total height started to decelerate, and a peak in pressure was observed simultaneously (Supporting Information 5); these features were typical of the crust.[41]

Local Compression in the Outer Layers of Dough. At 7 min, the bread was composed of open and closed pores (Figure 3d). Where pores were closed, gas pressure was high and where pores were open, gas pressure was near atmospheric pressure (Figure 3e). Where pores were open, pressure forces did not counterbalance the gravity forces, and therefore inflation was not possible. At closed porosity (in the centre region), gas pressure was higher than gravity stress, therefore porosity increased in the centre (Figures 10a2, b2, c2, and d2).

Still at 7 min of baking, the top crust temperature was close to 120°C (Figure 3a) and moisture content was less than 0.1 kg kg_{dm}^{-1} (Figure 3b), which resulted in high viscosity (2×10^7 Pa.s—Figure 3c). This created a new stress which tended to stop the dough expansion overall. As the centre was in course of inflation, the rest of the dough tended to be squeezed by compression between two areas, one of high pressure (Figures 3e–10b2) and the other of high viscosity (Figures 3c–

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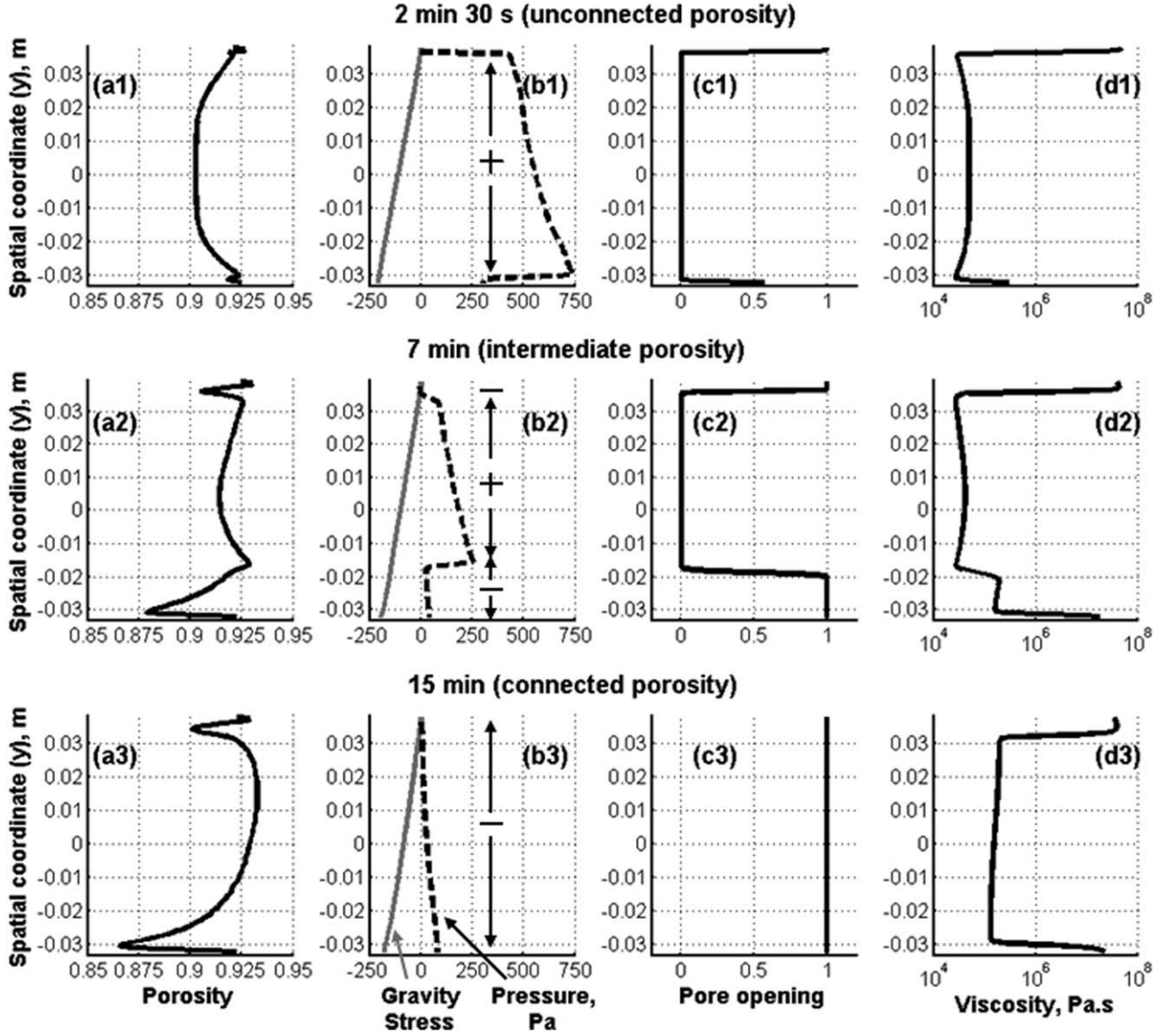


Figure 10. Porosity (column a), Gravity stress (solid grey line), and gas pressure relative to the atmospheric pressure (dashed line) (column b), pore opening function (column c), dynamic viscosity (column d).

Profiles along the line of symmetry at 2.5 (line 1), 7 (line 2), and 15 min (line 3) of baking for the reference simulation; "+" and "−" signs delimit the areas where pressure is higher or lower than gravity stress, respectively.

10d2). At this moment, the region of high viscosity was not thick enough to hold the overall structure, explaining why total height continued to decrease (Figure 8a). Temperature and moisture dependence of the viscosity model was fundamental to creating this effect with simulation. Models of baking implemented for two dimensions and using only temperature-dependent mechanical properties failed to simulate this kind of behavior.[3]

This phenomenon also applied to dough layers near the bottom surface. The parametric study showed that gravity also enhanced compression in this area (Figure 11b). This was consistent with the force balance presented in Figure 10. While compression occurred in top and bottom regions, the main explanation for why porosity was lower in the bottom region was gravity. To some extent, the additional compression caused by gravity in the lower region even decreased the level of squeezing in the upper parts (Figures 11a, b).

Decrease in Total Height. A greater decrease in total height was observed with gravity (Figure 8b). Gravity contributed to two-thirds of this decrease at 30 min of baking, shrinkage by dehydration being responsible for the other one-third (discussed further below). A slight decrease in relative height was observed even without gravity (Figure 8b). Shrinkage was the only phenomenon that could explain this behavior. Prolonged drying in the outer dough layers also reduced the porosity in neighboring inner regions by shrinkage (Figure 11c, top region). The simulation without gravity highlighted the need to take it into account to simulate bread deformation during baking properly, both overall and locally.

Wagner et al.[42] showed that inflation could proceed locally, well after inflation of the whole dough had ceased. This study showed that a decrease in total height could occur while inflation was still proceeding locally (Figure 8a vs. Figures 9b, c showing experimental inflation up to 10 min). The reason for this was the predominance of local compression by gravity

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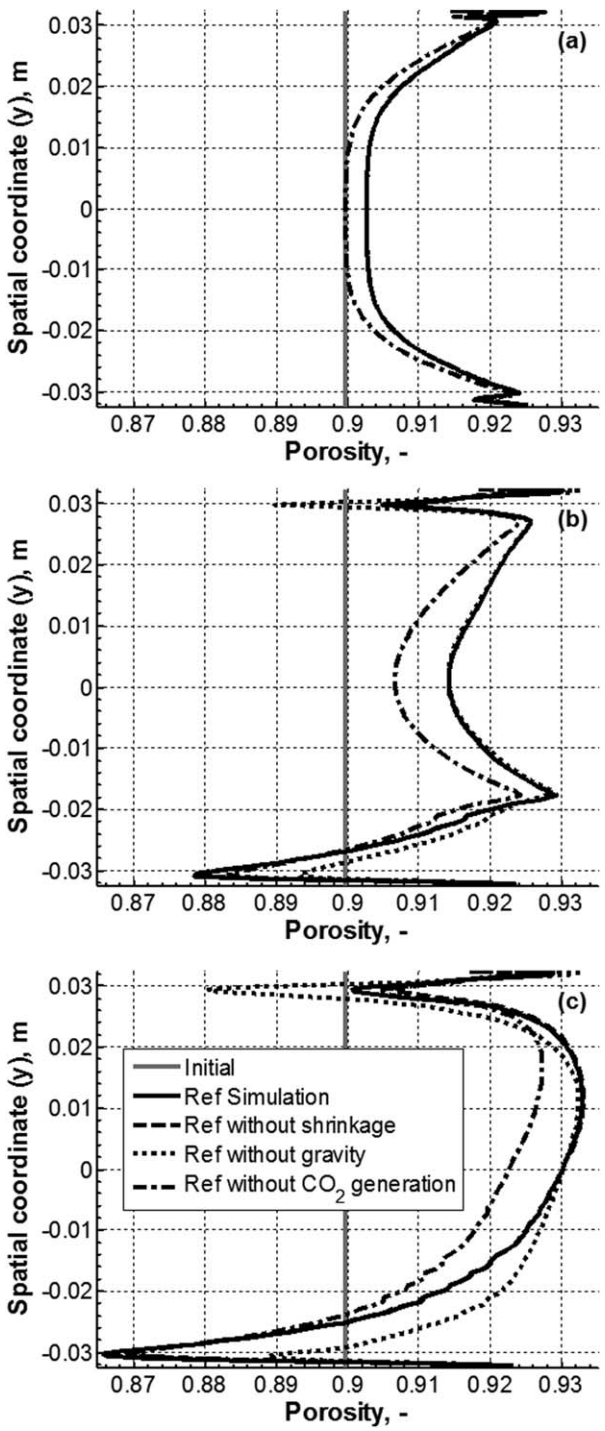


Figure 11. Porosity profiles along the line of symmetry at 2.5 (a), 7 (b) and 15 min of baking (c).

Note that at 2.5 min of baking, most of the dough section remained close to the temperature of the proving chamber, especially at the core (Figure 4, also Supporting Information 3 and 4) where the amount of CO_2 was nevertheless increasing.

over local inflation, together with a still deformable outer layer of dough, although at a slow rate. From 5 min of baking, the pores connected to the outside occupied a region large enough for this net decrease in total height (Figure 8a).

Height Stabilization at Prolonged Baking Times. The pressure decrease in the region of pore opening (Figure 10b2) was followed by a slight increase in pressure (Figure 10b3, also Supporting Information 5) due to water boiling, observed from 15 min of baking when the core temperature was close to 100°C (Figure 4). The air phase which penetrated at earlier baking times (Supporting Information 3, 15 min) was released outside the product and replaced by water vapor (data not shown). The pressure profile along the line of symmetry was then linear, with the highest value beside the mould and the atmospheric pressure at the upper surface (Figure 10b3). The slope of this profile is related to the permeability value. During the end of baking (after 20 min), this pressure profile was lower than that of the gravity stress, but only slightly, thus explaining why the total height decreased at a much slower rate than at earlier times. Moreover, the level of gas pressure due to water boiling was low compared to that corresponding to the first period of pressure inflation resulting from thermal expansion of gases and release of carbon dioxide from the liquid phase to the gaseous phase. This also contributed to the stabilization of total height at the end of baking. These results showed that water vapor impacted on the deformation processes very late in the baking process, with low dynamics, whereas carbon dioxide governed the main changes in porosity occurring during oven-rise.

Crust thickness was also a factor contributing to the stabilization of the total dough volume. During baking, top crust thickness increased with the loss of water (about 4 mm after 15 min, and 10 mm after 45 min). The product was therefore enclosed by a thicker and thicker ring of solid matter (high viscosity—Figure 10d3) which tended to immobilize the whole structure (Supporting Information 3 and 4). This phenomenon was possible because of the 2D geometry.

Conclusion

This two-dimensional numerical model, validated overall for temperature and mass evolution, allowed focusing on overall and local porosity which for the first time were validated experimentally in a complete baking process. The present work takes into account many coupled phenomena in a 2D geometry, notably evapo-condensation even for unconnected pores, presence of air in the gas phase and presence of CO_2 in the liquid phase. Moreover, for the first time, the rheology is considered to depend both on temperature (direct influence and influence through starch gelatinization and gluten reticulation) and water content. Additionally shrinkage due to water loss was included in the mechanical model. Experimental changes in porosity in the different dough regions were reproduced very satisfactorily by the baking model, with net progress compared to unidirectional study[2] which revealed discrepancies between simulation and experiment in the squeezed regions.

A parametric study highlighted that carbon dioxide generation by yeast during baking constitutes a minor contribution compared to dissolved and gaseous amounts of CO_2 present at the end of the proving step; surprisingly, it also showed that its effect on local and overall expansion was far from negligible (-42% of final relative height without CO_2 generation). This contribution may be higher with boosted yeasts or other chemical reaction- producing leavening agents.

To increase local porosity, pressure forces must overcome gravity forces. The analysis of these forces provided an explanation for how high pressure in the centre in conjunction with a low rate of deformation in the outer layers at intermediate baking times (high viscosity 2×10^7 Pa.s) induced local compression in the outer dough layers. The regions being squeezed

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were those presenting the lowest pressure levels. Gravity enhanced the compression phenomenon at the bottom where gravity was highest, which in turn lessened the squeezing phenomenon at the top.

Gravity contributed two-thirds of the decrease in total height during baking (calculated at 30 min of baking), shrinkage by dehydration being responsible for the other third. The changes in local porosity by gravity were small (maximum 0.025), but located in a wide area at the bottom (3 cm thick). In this study, dough stiffening occurred just after pore opening. The effects of gravity at the product level can either be enhanced for weaker flour or starch with delayed gelatinization, or greatly reduced for stronger flour than that used in this study. The effects of shrinkage during baking were hardly noticeable at the local level. This study also showed that maximum inflation was not related to the crust setting process, but to the greater collapse of certain pores than the inflation of other pores.

The stabilization of total height at the end of baking was explained by a pressure increase due to water boiling at the core of the product which counterbalanced the effect of gravity, and by increasing thickness of the crust, being defined as the area of slow deformability (high viscosity).

In further work, the post baking cooling step will be simulated. Indeed mass transfer and deformation continue to occur during this period. Therefore to predict the final product characteristics, cooling should be included in the simulations. The model will also be used to analyse different operating conditions of baking: varying air temperature and humidity, air circulation, radiation in the oven. To predict realistically the texture of the bread, a visco-elastic model (as used in Zhang et al.[5] or Nicolas et al.[3]) should be used instead of a simple viscous model. Although poorly understood, pore opening also affects the changes in the porous structure. Consistently with the state of the art, a temperature criterion was retained for pore opening. Future research would be required to make this parameter more physically-consistent, relating it to the local mechanics (level of strain?) or even the mass transport at the micro scale as starch gelatinises.

Notation

Greek letters

α = pore opening function
δ = ratio of film thickness to total film and bubble thickness
ΔT = pore opening temperature range, K
ε = porosity, $m^3_{(l+g)}\ m^{-3}_{product}$
$\varepsilon_l, \varepsilon_g, \varepsilon_s$ = liquid, gas, and solid volume fraction, $m^3\ m^{-3}_{product}$
η = dynamic viscosity, Pa s
λ = thermal conductivity of product, W m^{-1} K^{-1}
ν = kinematic viscosity of gas, m^2 s^{-1}
ξ = boundary mass transfer coefficient, m^{-1} s^{-1}
ρ = density, kg m^{-3}
$\overline{\overline{\sigma}}_{sh}$ = shrinkage stress, Pa
$\overline{\overline{\tau}}, \tau$ = viscous stress, Pa; tortuosity
ω = mass fraction (wv, cv, a) in the gaseous phase, kg kg^{-1}

Letters

a_w = water activity
A_{wvvc} = evaporation-condensation diffusion coefficient, s
c_p = specific heat, J kg^{-1} K^{-1}
D = diffusion coefficient, m^2 s^{-1}
$\vec{g}$ = gravity acceleration, m s^{-2}
G_T, G_{Xwl} = contribution of thermal and moisture content contribution to dynamic viscosity
h = convection heat transfer coefficient, W m^{-2} K^{-1}
$\dot{H}$ = volume enthalpy, J m^{-3}
$\vec{H}$ = enthalpy flux, J m^{-2} s^{-1}
H_c = Henry's law constant
$\overline{\overline{I}}$ = identity matrix
I_c = carbon dioxide generation, kg of CO_2 m^{-3} of total volume s^{-1}
k_{conv} = convection mass transfer coefficient, m s^{-1}
K = gas permeability, m^2
L = latent heat of vaporization, J kg^{-1}
Le = Lewis number
$\vec{m}$ = mass flux, kg m^{-2} s^{-1}
M = molar mass, kg mol^{-1}
P = pressure, Pa
$\vec{Q}$ = conduction heat flux, J m^{-2} s^{-1}
R = ideal gas constant, J mol^{-1} K^{-1}
r_0 = initial dough radius, m
R_{bubble} R_{film} = evaporation-condensation resistance, s^{-1}
S = triangular surface between oil microcapsule spots as observed in the MRI image, m^2
T = temperature, K
T_{open} = pore opening temperature, K
t = time, s
$\vec{u}$ = velocity of dry matter, m s^{-1}
X = mass content relative to dry matter, kg kg_s^{-1}

Subscript

0 = initial
a = air phase (without carbon dioxide and water vapor)
atm = atmosphere
b = boundary
md = mould
cont = contact with mould
conv = convection
c = carbon dioxide
d, db = Darcy, Darcy boundaries
di = diffusion
dgh = dough without gas (continuous phase)
g = gas (water vapor + carbon dioxide vapor + air)
l = liquid (water) or dissolved (carbon dioxide) state
oven = oven
rad = radiation
ref = reference
s = solid
sat = saturated
surf = surface
wvvc = evaporation-condensation
v = vapor state
w = water

Superscript

a = apparent
eff = effective
i = intrinsic

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Appendix: Resolved Equations

Solid volume fraction equation, ε_s

$$\rho_s^i \frac{\partial \varepsilon_s}{\partial t} + \vec{\nabla} \cdot \left(\rho_s^i \varepsilon_s \vec{u}\right) = 0 \tag{A1}$$

Liquid moisture content equation, X_{wl}

$$\rho_s^a \frac{\partial X_{wl}}{\partial t} + \vec{\nabla} \cdot \left(\vec{m}_{wl} + \vec{m}_{wvv} + \rho_s^a (X_{wv} + X_{wl}) \vec{u}\right) = -\left((X_{wl} + X_{wv}) \frac{\partial \rho_s^a}{\partial t} + \rho_s^a \frac{\partial X_{wv}}{\partial t}\right) \tag{A2}$$

Liquid carbon dioxide content equation, X_{cv}

$$\rho_s^a \frac{\partial X_{cl}}{\partial t} + \vec{\nabla} \cdot \left(\vec{m}_{cl} + \vec{m}_{cv} + \left(\rho_s^a X_{cv} + \rho_s^a X_{cl}\right) \vec{u}\right) = -\left(\rho_s^a \frac{\partial X_{cv}}{\partial t} + (X_{cv} + X_{cl}) \frac{\partial \rho_s^a}{\partial t}\right) + I_c \tag{A3}$$

Gas pressure equation, P

$$\frac{M_a \varepsilon_g}{RT} \frac{\partial P}{\partial t} + \vec{\nabla} \cdot \left(\vec{m}_a + \rho_a^a \vec{u}\right) = \frac{M_a \varepsilon_g}{RT} \frac{\partial P_{wv}}{\partial t} + \frac{M_a \varepsilon_g}{RT} \frac{\partial P_{cv}}{\partial t} + \frac{M_a \varepsilon_g}{RT^2} (P - P_{wv} - P_{cv}) \frac{\partial T}{\partial t} - \frac{M_a}{R} \frac{(P - P_{wv} - P_{cv})}{T} \frac{\partial \varepsilon_g}{\partial t} \tag{A4}$$

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Displacement velocity equation

$$\rho\frac{\partial\vec{u}}{\partial t}+\vec{\nabla}\cdot\begin{pmatrix}\begin{pmatrix}-\frac{4}{3}\eta\frac{\partial u}{\partial x}+\frac{2}{3}\eta\frac{\partial v}{\partial x}\\ +P-P_0+\left[\alpha\frac{\partial X_{wl}}{\partial t}\right]+\rho uu\end{pmatrix} & -\eta\left(\frac{\partial u}{\partial y}+\frac{\partial v}{\partial x}\right)+\rho uv\\ -\eta\left(\frac{\partial u}{\partial y}+\frac{\partial v}{\partial x}\right)+\rho vu & \begin{pmatrix}-\frac{4}{3}\eta\frac{\partial v}{\partial y}+\frac{2}{3}\eta\frac{\partial u}{\partial x}\\ +P-P_0+\left[\alpha\frac{\partial X_{wl}}{\partial t}\right]+\rho vv\end{pmatrix}\end{pmatrix}=-\vec{u}\frac{\partial\rho}{\partial t}+\rho\vec{g} \tag{A5}$$

Heat equation

$$\begin{aligned}\rho_s^a c_p\frac{\partial T}{\partial t}+\vec{\nabla}\cdot\left(\vec{\dot{Q}}+\vec{\dot{H}}+\hat{H}\vec{u}\right)&=-\rho_s^a\left(c_{p;wv}\left(T-T_{ref}\right)+L_w\right)\frac{\partial X_{wv}}{\partial t}\\&-\rho_s^a c_{p;wl}\left(T-T_{ref}\right)\frac{\partial X_{wl}}{\partial t}-\rho_s^a\left(c_{p;cv}\left(T-T_{ref}\right)+L_c\right)\frac{\partial X_{wc}}{\partial t}\\&-\rho_s^a c_{p;cl}\left(T-T_{ref}\right)\frac{\partial X_{cl}}{\partial t}-\rho_s^a c_{p;a}\left(T-T_{ref}\right)\frac{\partial X_a}{\partial t}\\&-\left(L_{H_2O}X_{wv}+L_{CO_2}X_{cv}+c_p\left(T-T_{ref}\right)\right)\frac{\partial\rho_s^a}{\partial t}\end{aligned} \tag{A6}$$

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